

Fold3D: Rethinking and Parallelizing Computational and Communicational Tasks in the Training of Large DNN Models (IEEE TPDS 23)

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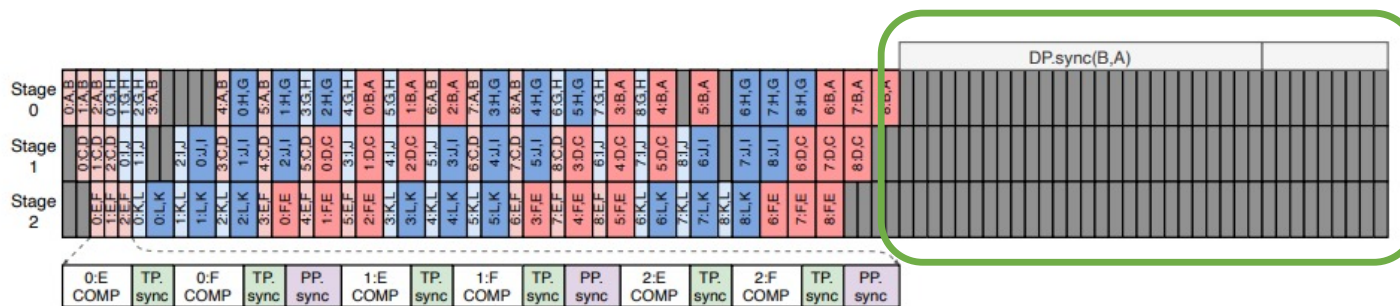
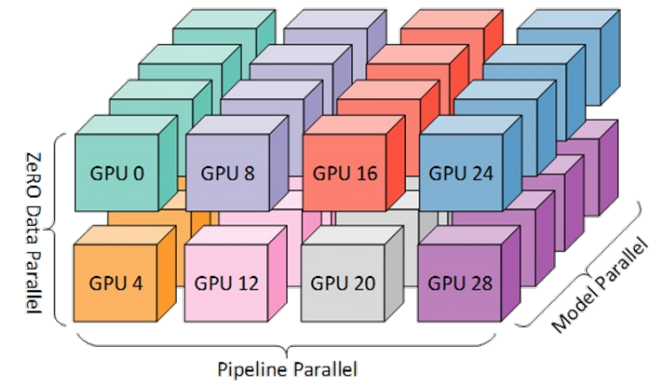
The University of Hong Kong



Huawei Technologies

Motivation of Fold3D

- 3D parallelism is essential for large DNN training
- To hold model: Tensor Parallelism (TP) and Pipeline Parallelism (PP)
 - TP within a host (e.g., DGX with NVLink)
 - PP across hosts (e.g., RDMA NICs)
- To accelerate: Data Parallelism (DP)
- Problem: DP communication serialized after PP computation
 - E.g., DP takes more than 40% step time on public cloud

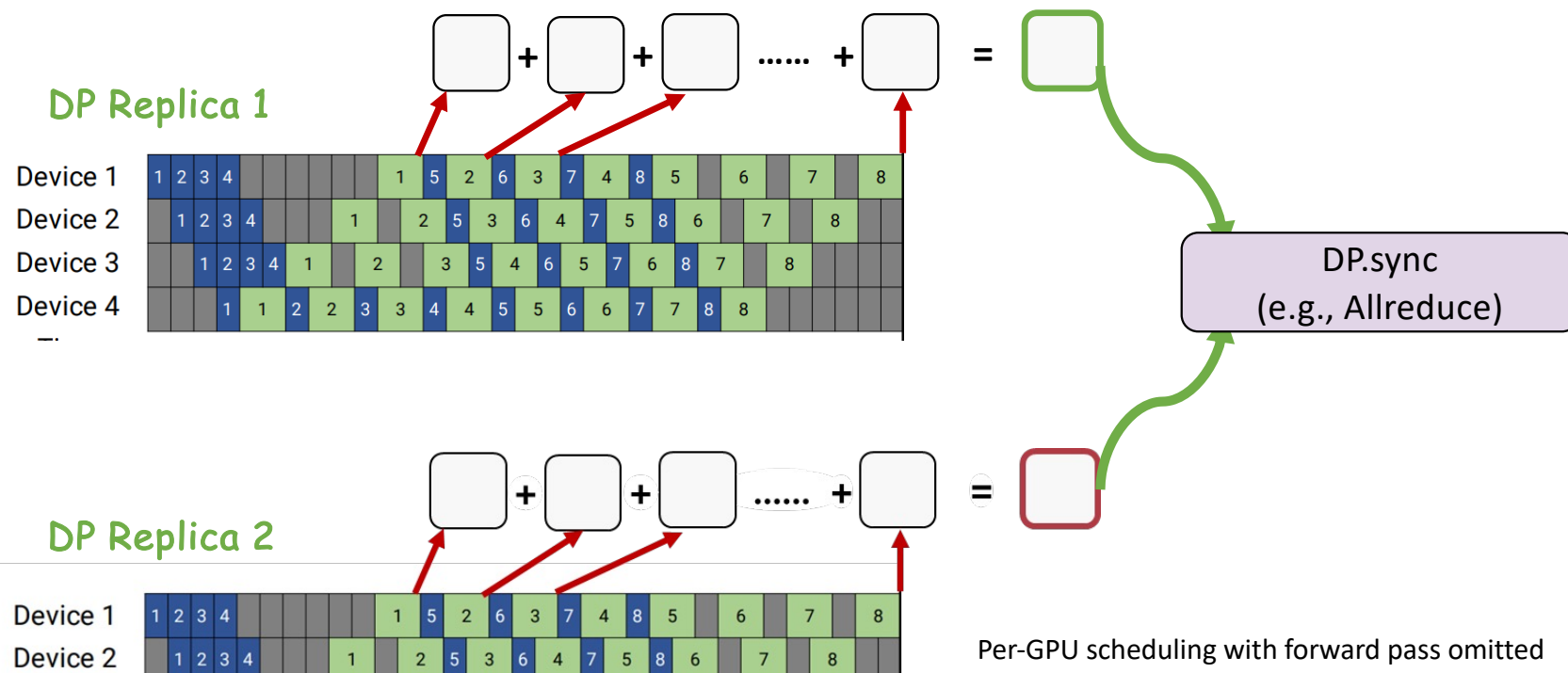


a) Megatron's 1F1B Scheduling

Data Parallel Allreduces (DP.sync) Causes GPUs Idle

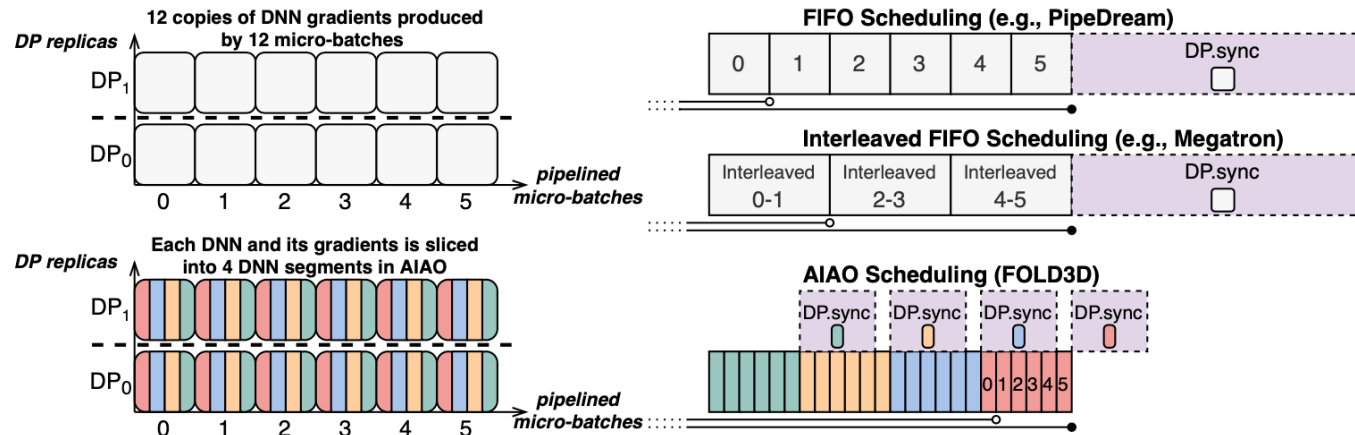
Root Rationales

- Pipeline Parallelism: Repeated-Microbatches with Gradient-Accumulation
- Data Parallelism: Replication of Computation and Sync of Accumulated Gradients



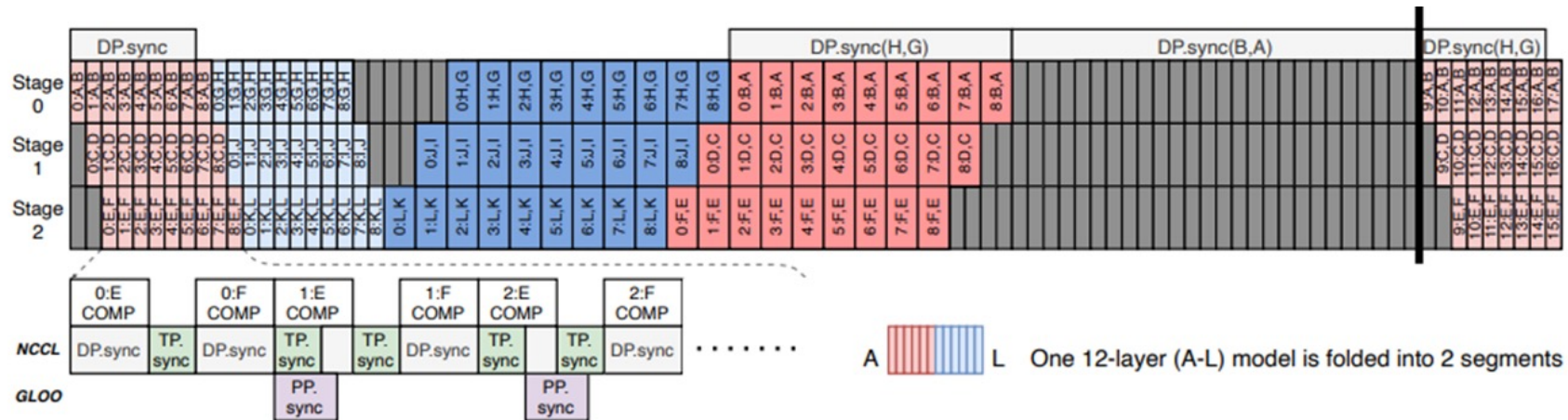
Idea - AIAO instead of FIFO

- Current 3D systems: First-In-First-Out (FIFO) in terms of micro-batches
- Fold Scheduling: All-In-All-Out (AIAO) in terms of micro-batches
 - Split gradient's computation, and group gradients by microbatch
 - Thus, DP.sync of the grouped gradients can be overlapped with other groups' computation



The Fold Scheduling

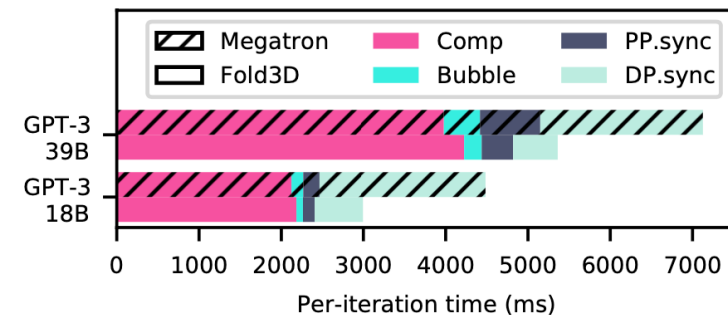
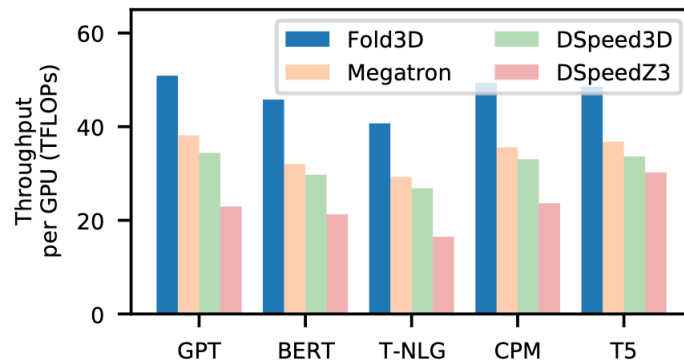
- DNNs are split into segments
 - E.g., 12-layer (A-L) DNNs is split to 2 segments over 3 stages
- The same segment of all micro-batches are grouped together
 - E.g., micro-batch 0-8 of Seg(A, B) are grouped together
- DP.sync of a segment can be overlapped with other segments
 - E.g., the DP all-reduce of (H,G) can be overlapped with the execution of Seg (A, B)



b) FOLD3D's AFAPAB Scheduling

Evaluation of Fold3D

- Fold3D significantly outperforms the baselines
 - PyTorch + 256 A100 GPUs: 31%+ throughput Improvement v.s. Megatron 3.0
 - MindSpore + 64 Ascend 910 NPUs: 30%+ v.s. MindSpore
 - The communication not overlapped with computation of Fold3D is less than that of Megatron-PTD



Industry Impact (1/2): Fold3D

• Ascend Speed: open-source industry-grade LLM training system

2023/10/6 23:18

Ascend/AscendSpeed



开源软件 企业版 高校版 私有云 博客



Ascend / AscendSpeed

代码 Issues 3 Pull Requests 9 Wiki 统计 流水线 服务

加入 Gitee

与超过 1000 万 开发者一起发现。参与优秀开源项目，私有仓库也完全免费：

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已有帐号？ 立即登录

master 分支 2 标签 0

文件

Web IDE

克隆/下载

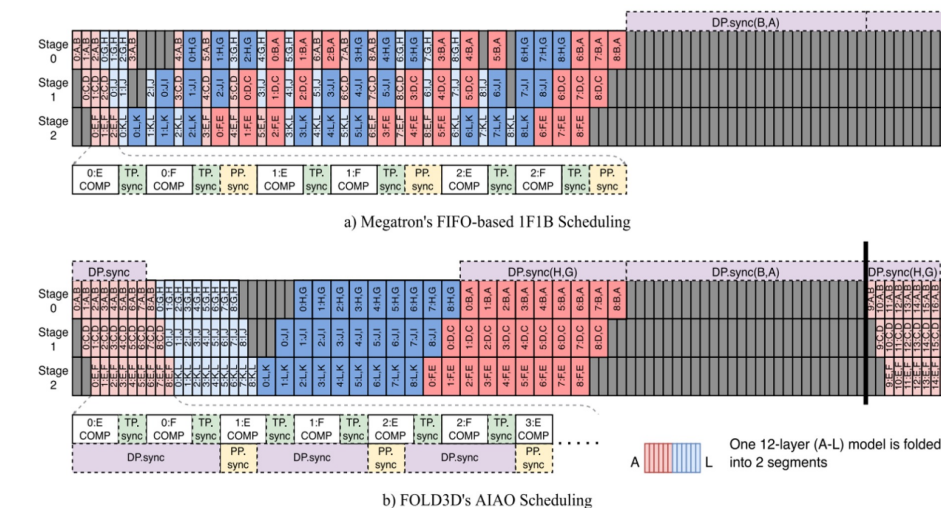
i-robot	I91change llama65B batch size	20e090b	8天前	63 次提交
ascendspeed	baichuan13B模型			9天前
ci	update ci/access_control_test.py.			10天前

- Tensor Parallelism
- (Virtual & Opti...
- Fold3D
- (Selective) Reco...
- Sequence Parall...
- ZeRO-1/2/3
- Inverted Triangl...
- Fused Kernels &...
- Merged Feed-Fo...
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- Preprocessing in...
- alpaca dataset
- Finetune with Lora

config-1 flag based on PP. Note that both VP and OPP reduce bubble time, but increase communication time.

Fold3D

Fold3D hides the commutation time of data parallelism in VP. The basic principle of Fold3D is:



To enable pipeline model parallelism, use the `--fold-mode "aiao"` flag to choose strategy.

Industry Impact (2/2): Fold3D

- Fold3D has been integrated into Huawei's MindSpore
- https://github.com/mindspore-ai/mindspore/blob/master/mindspore/ccsrc/frontend/parallel/pipeline_transformer/fold_pipeline_transformer.cc

```
37  class FoldPipelineTransformer : public PipelineTransformer {
38      public:
39          FoldPipelineTransformer(const FuncGraphManagerPtr &manager, int stage, const FuncGraphPtr &root, int64_t
40                                  int64_t per_stage_rank_num)
41              : PipelineTransformer(manager, stage, root, global_rank, per_stage_rank_num) {}
42          ~FoldPipelineTransformer() = default;
43          void Coloring() override;
44          void BroadCastColoring() override;
45          void CutGraph() override;
46
47          SendAttr InsertSend(const AnfNodePtr &parameter, int64_t user_node_stage, int64_t node_stage, const Val
48                              int64_t segment);
49          AnfNodePtr InsertReceive(const FuncGraphPtr &graph, const AnfNodePtr &node, const AnfNodePtr &use_node,
50                                  int64_t + user node stage, int64_t + node stage, const ValuePtr &value.
```

```
mindspore / mindspore / ccsrc / frontend / parallel / pipeline_transformer / fold_pipeline_transformer.cc
Code Blame 709 lines (674 loc) · 29.9 KB Raw
40 #include "ops/framework_ops.h"
41 #include "include/common/utils/comm_manager.h"
42 #include "utils/ms_context.h"
43 #include "utils/parallel_node_check.h"
44
45 namespace mindspore {
46     namespace parallel {
47         mindspore::HashMap<int64_t, int64_t> fold_send_tag_map;
48         mindspore::HashMap<int64_t, int64_t> fold_recv_tag_map;
49
50         void FoldPipelineTransformer::CreateForwardGroup2() {
51             auto rank_id = g_device_manager->global_rank();
52             auto stage_id = g_device_manager->stage_id();
53             auto stage_num = g_device_manager->stage_num();
54
55             std::vector<int64_t> forward_rank_list;
56             forward_rank_list.push_back(rank_id);
57             if (stage_id < stage_num - 1) {
```